

المخبر 20 | LAB 20

# Autonomous Microgrid Supervisor

مشرف ذاتي التشغيل لشبكة مصغرة

Integrated MATLAB capstone with forecasting, battery scheduling, anomaly detection, and safety filtering

مشروع MATLAB متكامل للتنبؤ وجدولة البطارية وكشف الشذوذ ومرشح الأمان

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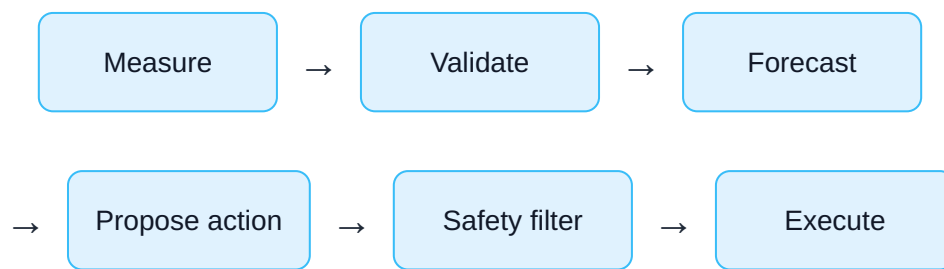
## Laboratory Objectives

- Simulate a 24-hour microgrid with load, PV, battery, and grid exchange.
- Forecast one hour ahead using a simple predictive model.
- Schedule battery actions using an economic policy.
- Detect abnormal measurements.
- Apply deterministic SOC and grid-power safety limits.
- Compare autonomous operation with a no-control baseline.

## أهداف المخبر

- محاكاة شبكة مصغرة لمدة 24 ساعة.
- التنبؤ بالساعة التالية.
- جدولة البطارية اقتصاديًا.
- كشف القياسات الشاذة.
- تطبيق حدود SOC وقدرة الشبكة.
- المقارنة مع خط أساس.

# 1. Integrated Control Loop



## 1. حلقة التحكم المتكاملة

تبدأ بالقياس والتحقق ثم التنبؤ واقتراح القرار وفلترته قبل التنفيذ.

## 2. Complete MATLAB Script

```

%% Lab 20: Autonomous Microgrid Supervisor
clear; clc; close all;
rng(20);

%% 1) Time series
T = 24;
hour = (0:T-1)';

loadKW = [2.1 2.0 1.9 1.8 1.9 2.3 ...
          3.0 3.8 4.4 4.7 4.5 4.2 ...
          4.0 4.1 4.5 5.2 6.0 6.8 ...
          6.4 5.7 4.9 4.1 3.2 2.6]';

pvKW = max(0,5.5*sin(pi*(hour-6)/12));
pvKW = pvKW.*(0.90+0.08*randn(T,1));

price = [0.18*ones(6,1); ...
         0.25*ones(10,1); ...
         0.42*ones(5,1); ...
         0.23*ones(3,1)];

%% 2) Insert one abnormal sensor measurement
loadMeasured = loadKW + 0.08*randn(T,1);
loadMeasured(18) = loadMeasured(18)+4.5;

%% 3) Battery parameters
capacity = 12;          % kWh
pMax = 3.0;             % kW
etaC = 0.95;
etaD = 0.95;
socMin = 0.10;
socMax = 0.90;
soc = zeros(T+1,1);
soc(1) = 0.50;

gridLimit = 7.0;        % kW
batteryPower = zeros(T,1);
gridPower = zeros(T,1);
forecastLoad = zeros(T,1);
alarm = false(T,1);
cost = zeros(T,1);

%% 4) Baseline cost without battery
baselineGrid = max(0,loadKW-pvKW);
baselineCost = sum(baselineGrid.*price);

```

```

%% 5) Autonomous loop
for k = 1:T

    % A) Measurement validation
    if k >= 4
        recent = loadMeasured(k-3:k-1);
        mu = mean(recent);
        sigma = std(recent);

        if sigma < 0.05
            sigma = 0.05;
        end

        zScore = abs(loadMeasured(k)-mu)/sigma;

        if zScore > 4
            alarm(k) = true;

            % Replace suspicious measurement with local estimate
            validatedLoad = mu;
        else
            validatedLoad = loadMeasured(k);
        end
    else
        validatedLoad = loadMeasured(k);
    end

    % B) One-step forecast
    if k == 1
        forecastLoad(k) = validatedLoad;
    else
        forecastLoad(k) = ...
            0.65*validatedLoad + ...
            0.35*loadMeasured(k-1);
    end

    netForecast = forecastLoad(k)-pvKW(k);

    % C) Economic action proposal
    if price(k) <= 0.20
        proposedPower = -pMax;      % charge
    elseif price(k) >= 0.40
        proposedPower = pMax;       % discharge
    else

```

```

        proposedPower = 0;
    end

    % D) Peak-support override
    if netForecast > gridLimit
        proposedPower = ...
            min(pMax,netForecast-gridLimit);
    end

    % E) Safety filter for SOC limits
    maxCharge = ...
        (socMax-soc(k))*capacity/etaC;
    maxDischarge = ...
        (soc(k)-socMin)*capacity*etaD;

    if proposedPower < 0
        safePower = -min(-proposedPower,maxCharge);
    else
        safePower = min(proposedPower,maxDischarge);
    end

    % F) Grid-limit safety filter
    rawGridPower = ...
        validatedLoad-pvKW(k)-safePower;

    if rawGridPower > gridLimit
        extraDischarge = ...
            min(gridLimit-rawGridPower+pMax, ...
                maxDischarge-safePower);

        extraDischarge = max(0,extraDischarge);
        safePower = safePower+extraDischarge;
    end

    batteryPower(k) = safePower;

    % G) Battery state transition
    if safePower < 0
        soc(k+1) = soc(k) + ...
            etaC*(-safePower)/capacity;
    else
        soc(k+1) = soc(k) - ...
            safePower/(etaD*capacity);
    end
end

```

```

        soc(k+1) = min(max(soc(k+1),socMin),socMax);

    % H) Executed grid power and cost
    gridPower(k) = max(0, ...
        validatedLoad-pvKW(k)-safePower);

    cost(k) = gridPower(k)*price(k);
end

%% 6) Performance
autonomousCost = sum(cost);
saving = baselineCost-autonomousCost;
peakReduction = max(baselineGrid)-max(gridPower);

fprintf("Baseline cost    = %.2f EUR\n",baselineCost);
fprintf("Autonomous cost = %.2f EUR\n",autonomousCost);
fprintf("Saving          = %.2f EUR\n",saving);
fprintf("Peak reduction  = %.2f kW\n",peakReduction);
fprintf("Detected alarms = %d\n",sum(alarm));

%% 7) Plots
figure;
plot(hour,loadMeasured,"o-",LineWidth=1.0); hold on;
plot(hour,loadKW,"k",LineWidth=1.4);
stem(hour(alarm),loadMeasured(alarm),"r",LineWidth=1.5);
grid on;
xlabel("Hour");
ylabel("Load (kW)");
legend("Measured","True","Alarm");
title("Measurement Validation");

figure;
stairs(0:T,soc,LineWidth=1.5);
grid on;
xlabel("Hour");
ylabel("SOC");
title("Battery State of Charge");

figure;
stairs(hour,batteryPower,LineWidth=1.4);
grid on;
xlabel("Hour");
ylabel("Battery Power (kW)");
title("Battery Action: Positive = Discharge");

```



```
figure;
plot(hour,baselineGrid,"--",LineWidth=1.2); hold on;
plot(hour,gridPower,LineWidth=1.5);
yline(gridLimit,"r:");
grid on;
xlabel("Hour");
ylabel("Grid Import (kW)");
legend("Baseline","Autonomous","Grid Limit");
title("Grid Import and Safety Limit");
```

## 2. الكود الكامل

ينفذ الكود حلقة ذاتية تشمل التحقق من القياسات والتنبؤ والقرار الاقتصادي ومرشح SOC وحد الشبكة والتنفيذ وتقييم الأداء.

## 3. Measurement Validation

$$z = |measurement - localmean| / localstandarddeviation$$

A suspicious measurement is replaced by a local estimate before control.

In real systems, anomaly detection should use multiple channels, topology, temporal context, and independent validation.

## 3. التحقق من القياسات

تُقارن القيمة الحالية بالمتوسط والانحراف المحليين، وتُستبدل القيمة الشاذة بتقدير مؤقت قبل اتخاذ القرار.

## 4. Forecasting Module

$$\hat{P}_{load}(k) = 0.65P_{validated}(k) + 0.35P_{measured}(k - 1)$$

This simple model is intentionally transparent. It can later be replaced by LSTM or transformer forecasting.

## 4. وحدة التنبؤ

يستخدم المثال نموذجًا بسيطًا شفافًا، ويمكن استبداله لاحقًا بـ LSTM أو Transformer.

## 5. Economic Policy

- Charge during low-price hours.
- Discharge during high-price hours.
- Support the grid when predicted import exceeds the limit.

## 5. السياسة الاقتصادية

تشحن البطارية عند السعر المنخفض وتفرغ عند السعر المرتفع وتدعم الشبكة عند تجاوز القدرة المتوقعة للحد.

## 6. Safety Filter

The proposed action is clipped by available energy and SOC limits before execution.

$$P_{safe} = Project(P_{proposed}, feasiblebatteryset)$$

## 6. مرشح الأمان

يُسقط القرار المقترح على مجموعة الأفعال الممكنة التي تحقق حدود SOC والطاقة والقدرة.

## 7. Performance Indicators

### Energy cost

Autonomous versus baseline cost.

### Peak reduction

Reduction in maximum grid import.

### Safety

Number of SOC or grid-limit violations.

### Anomaly detection

Detected suspicious measurements.

## 7. مؤشرات الأداء

تشمل الكلفة وخفض الذروة والمخالفات وعدد القياسات الشاذة المكتشفة.

## 8. Student Experiments

1. Replace the simple forecast with the LSTM from Lab 09.
2. Replace the rule-based policy with the DQN agent from Lab 12.
3. Add a GNN-based state-estimation module.
4. Simulate PV forecast error.
5. Simulate a communication outage.
6. Add a human-approval flag for high-impact actions.

## 8. تجارب الطالب

1. استبدل التنبؤ بـ LSTM.
2. استبدل السياسة بوكيل DQN.
3. أضف تقدير حالة باستخدام GNN.
4. أضف خطأ توقع PV.
5. حالك انقطاع الاتصال.
6. أضف موافقة بشرية للقرارات المهمة.

## 9. Capstone Extension

**Full autonomous-grid prototype:** Connect the forecasting, digital twin, anomaly detection, optimization, safety filter, and operator-assistant modules developed throughout the course.

## 9. مشروع ختامي

اربط وحدات التنبؤ والتوأم الرقمي وكشف الشذوذ والتحسين ومرشح الأمان ومساعد المشغل ضمن نموذج واحد.

## 10. Laboratory Report

- Draw the autonomous-loop architecture.
- Explain every module.
- Report cost and peak reduction.
- Analyze anomaly detection.
- List all safety constraints.
- Discuss human oversight and failure modes.

## 10. تقرير المخبر

يتضمن المعمارية وشرح الوحدات والكلفة وخفض الذروة وكشف الشذوذ وقيود الأمان والإشراف البشري وحالات الفشل.

# المراجع | References

المراجع العامة المستخدمة في هذا المقرر

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